Coefficient infinity-norm SVP on principal cyclotomic ideals

27 October 2026

1 Coefficient-infinity SVP on principal power-of-two cyclotomic ideals

Theorem 1. *For the fixed ring family $R_n = \mathbb{Z}[X]/(X^n+1)$, with n ranging over powers of two, the following problem is NP-complete under deterministic polynomial-time many-one reductions: given $f \in R_n \setminus \{0\}$ and an integer $r \geq 1$, decide whether $\lambda_{1,\infty}(fR_n) \leq r$, where the norm is the coefficient infinity norm. The input lists the n integer coefficients of f and encodes r in binary.*

The argument below is purely theoretical. The dimension grows along the predefined power-of-two family; it does not assert hardness in one constant numerical dimension. No conclusion for coefficient Euclidean norm, canonical infinity norm, or constant-factor approximation is inferred from this theorem. The reduction has an additive gap: YES instances have $\lambda_{1,\infty} \leq r$, while NO instances have $\lambda_{1,\infty} \geq r+2$. The radius can have exponentially large numerical value, so this does not yield a constant-factor approximation gap.

1.1 A source graph encoded by integer differences

We reduce from CLIQUE on a graph Γ with vertex set $\{1,\dots,v\}$ and a target $3 \leq h \leq v$. Put $b_0 = 3v^3+1$ and

$$a_i = i + b_0 i^2 + b_0^2 i^3, \qquad A_0 = \max_i a_i, \qquad D = \{|a_i - a_j| : ij \in E(\Gamma)\}.$$

These labels have the following three-term uniqueness property: equality of two sums of three labels, allowing repetitions, implies equality of the two index multisets. Indeed there is no carry in base b_0, so the first three power sums of the three indices agree; Newton's identities determine their monic cubic. In particular, nonzero ordered differences of labels are unique. All labels are polynomially bounded in v.

Lemma 1 (Difference-clique extraction)**.** *If an integer set P has at least three elements and every positive pair difference of P belongs to D, then its size is at most v and its elements have the form*

$$t + \varepsilon\{a_i : i \in C\}, \qquad \varepsilon \in \{-1,1\},$$

for a clique C of Γ.

Proof. Translate one element of P to zero. Every nonzero remaining element uniquely has the form $a_i - a_j$, with ij an edge. For two distinct such elements $a_i - a_j$ and $a_k - a_l$, their difference is $a_p - a_q$ for an edge pq. Three-term uniqueness therefore gives $\{i,l,q\} = \{j,k,p\}$ as multisets. If $i \neq k$, then $i \neq j$ implies $i = p$, whence $\{l,q\} = \{j,k\}$. Since $k \neq l$, this forces $l = j$. Thus any two representing ordered pairs share their first or their second coordinate. A family of at least two distinct pairs with this property lies in a single row or a single column: if two share their first coordinate but have distinct second coordinates, every other pair must share that first coordinate; the column case is symmetric. Their other indices, together with the common index, form a source clique, by uniqueness of ordered differences. This gives the asserted representation. □

1.2 A precise sparse-spin source

For $W \subseteq \{0, \dots, n-1\}$ of nonzero odd size, let $\sigma \in \{0, 1, -1\}^n$ have support W and alternate in sign on the increasing list of its support. Its initial sign is irrelevant. Extend it antiperiodically by $\sigma_{i+n} = -\sigma_i$. For an integer multiplier $G \in R_n$, put

$$\Theta(G) = \max_{\substack{W \neq \emptyset \\ |W| \text{ odd}}} \min_{i \in W} \sigma_i (G\sigma)_i.$$

The multiplication matrix and all inner products below refer to the coefficient basis. For $X^* = X^{-1}$, a skew-adjoint multiplier satisfies $\langle z, Az \rangle = 0$ for real vectors z.

Fix $Q = 16$ and write $R_0 = QA_0 + 2$. Choose n to be the least power of two greater than $8R_0 + 16$. In particular $n = \mathrm{poly}(v)$. Define the allowed positive distances

$$E_+ = \{1, 2\} \cup \bigcup_{d \in D} \{Qd - 1, Qd, Qd + 1, Qd + 2\}.$$

All these distances are at most $R_0 < n/8$. Allowed cyclic differences are exactly $E_+ \cup (-E_+)$ modulo n. The positive windows are formed only for positive $d \in D$; their negative windows are their reflections. In particular, no five-point window is introduced by symmetrizing its center.

List the forbidden unsigned cyclic distances as

$$\{d_0, \dots, d_{s-1}\} = \{1, \dots, n/2\} \setminus E_+, \qquad A = \sum_{j=0}^{s-1} 5^j (X^{d_j} - X^{-d_j}).$$

Define the self-adjoint small interaction

$$T = \sum_{d \in D} (X^{Qd} + X^{-Qd} + X^{Qd+2} + X^{-Qd-2}), \qquad R_T = \|T\|_1,$$

and choose the explicit integers

$$C = R_T + h + 1, \quad \tau = C + h - 1, \quad L = 2C + R_T + \tau + 1, \quad G = LA - C(X + X^{-1}) + T.$$

Lemma 2 (Forbidden-distance guard)**.** *If an odd alternating support σ satisfies $\sigma_i (G\sigma)_i \geq \tau$ on its support, then every pair of its support positions has an allowed cyclic difference.*

Proof. At a support coordinate where the integer $\sigma_i (A\sigma)_i$ were negative, the whole field would be at most $-L + 2C + R_T < \tau$. Thus all these skew fields are nonnegative. Their sum is $\langle \sigma, A\sigma \rangle = 0$, so all are zero, and hence $(A\sigma)_i = 0$ on the support. For each such coordinate,

$$\sum_j 5^j ((X^{d_j} - X^{-d_j})\sigma)_i = 0.$$

Each digit is an integer in $[-2, 2]$; uniqueness of balanced base-five representation makes every digit zero.

It remains to verify that zero active fields of $A_d = X^d - X^{-d}$ exclude a support pair at distance d. The equation is $\sigma_{i-d} = \sigma_{i+d}$ at every support coordinate. If one step-d pair were nonzero, these equations would propagate nonzero values throughout that step-d orbit, and would make them periodic with step $2d$. The orbit length $\ell = n/\gcd(n, d)$ is an even power of two, and ℓd is an odd multiple of n. Iterating the period $\ell/2$ times contradicts antiperiodicity. This applies also to $d = n/2$. Thus no forbidden support pair exists. □

1.3 The forced support geometry

Lemma 3 (Exactly one trimer)**.** *A support satisfying the preceding field inequalities consists of dimers (runs of length two) and exactly one trimer (a run of length three). All runs lie in one short integer interval. After cutting at the long empty circular gap, let the trimer start at t. Every dimer to its left starts at $t + Qq$ and every dimer to its right at $t + Qq + 1$, for some integer q. The set of these integers q, together with 0 for the trimer, has every positive pair difference in D.*

Proof. The forbidden guard makes the skew fields zero, so all active fields reduce to those of $-C(X + X^{-1}) + T$. Adjacent active signs are opposite. A support point with no adjacent support neighbor therefore has field at most $R_T < \tau$, and cannot occur. Since distance three is forbidden, every run has length two or three. A one-zero gap would put the last point of one run at distance three from the second point of the next; a two-zero gap would put the two endpoints at distance three. Thus gaps have at least three zeros.

Every pair of support points has circular distance at most R_0. To see that they lie in a short interval, move one to zero and represent the others in $[-R_0, R_0]$. The difference of the two extreme representatives is at most $2R_0 < n - R_0$, so its circular distance can be at most R_0 only if that difference itself is at most R_0. Thus the entire support lies in an integer interval of length at most R_0. The complementary circular gap has length at least $n - R_0$, giving a consistent linear ordering. Multiplication by a monomial X^k is a signed cyclic permutation which commutes with every multiplier. It preserves each local field under the induced permutation: if a coordinate changes sign, both factors in $\sigma_i(G\sigma)_i$ change sign. We may therefore place the entire short arc in $[0, R_0]$. All interactions between support positions can now be computed on this integer interval, without a negative-cyclic boundary crossing.

Two distinct trimers, whose start difference is $\delta \geq 6$, would supply all five distances $\delta - 2, \ldots, \delta + 2$. They are larger than two and hence must lie in far windows. Such windows have length four and are separated by at least $Q - 3 > 1$, so they cannot contain five consecutive distances. There is at most one trimer; odd support size forces exactly one.

For a left dimer with start $p < t$, its differences to the trimer are the four consecutive values $(t - p) + \{-1, 0, 1, 2\}$. These fit an allowed far window only if $t - p = Qd$ for some $d \in D$. For a right dimer with start $p > t$, those four values are $(p - t) + \{-2, -1, 0, 1\}$, forcing $p - t = Qd + 1$. Thus the asserted start positions follow. For two dimers on the same side, their three pairwise distance values are $Q|q - q'| + \{-1, 0, 1\}$; these lie in a window only if $|q - q'| \in D$. For two on opposite sides the values are $Q(q' - q) + \{0, 1, 2\}$, with the same conclusion. This proves the last assertion. □

Conversely, take any source clique, select one label for the trimer, place the other components according to the start rules in the lemma, and make the remaining components dimers. All component differences then fit the prescribed windows. Their separations are at least $Q - 2$, and they fit within an interval of length at most $QA_0 + 3 < n/2$. This constructs an admissible odd support with exactly the desired number of components.

1.4 The endpoint interaction table

At a trimer center the nearest-neighbor contribution is $2C$, so its whole small field is at least $2C - R_T > \tau$. At each run endpoint that contribution is C; it remains to compute $\sigma_i(T\sigma)_i$ there.

Use coordinates relative to the trimer start, retaining the integer lifts in the short interval, and choose its signs as $(+, -, +)$. A left dimer has signs $(+, -)$ and start Qq, while a right dimer has signs $(-, +)$ and start $Qq + 1$. Around a positive edge difference center Qd, the four coefficients of T are

$$(t_{Qd-1}, t_{Qd}, t_{Qd+1}, t_{Qd+2}) = (0, 1, 0, 1),$$

with self-adjoint reflection at negative distances. Contributions from distinct components are as follows:

receiver and sender	receiver endpoint contributions
dimer from another dimer, on either side	$(1,1)$
trimer from any dimer	$(1,1)$
left dimer from trimer	$(2,1)$
right dimer from trimer	$(1,2)$

For example a left dimer to the trimer's left has start $-Qd$. Its first endpoint receives $t_{Qd} - t_{Qd+1} + t_{Qd+2} = 2$, while its second endpoint receives $-t_{Qd-1} + t_{Qd} - t_{Qd+1} = 1$. The other rows follow by the same two- or three-term sums. There are no within-component contributions from T.

If there are q components, the trimer's two endpoint fields from T are exactly $q-1$. Every dimer has endpoint fields $q-1$ and q. Consequently

$$\min_{i \text{ a run endpoint}} \sigma_i(T\sigma)_i = q - 1.$$

The small-field inequalities hold exactly when $q \geq h$. Combining the forbidden guard, the geometric classification, the difference-clique extraction, and the converse construction, we have proved the exact decision equivalence

$$\boxed{\Gamma \text{ has an } h\text{-clique} \quad \Longleftrightarrow \quad \Theta(G) \geq \tau.} \tag{1}$$

1.5 From sparse spins to full signs and to a principal ideal

For completeness, define the full-sign local-field objective

$$\Psi(K) = \max_{s \in \{-1,1\}^n} \min_i s_i(Ks)_i.$$

Write $S = 1 + X + \cdots + X^{n-1}$. Put $H = G(1-X)$, $R_H = \|H\|_1$, $B = R_H + 2\tau + 1$, and $K = BX + H$. For a full sign vector s, set $e = (1-X)s$. The support of e consists of its negative-cyclic domain walls. Their number is odd, since $\prod_i s_i s_{i-1} = -1$. At these walls $e_i = 2s_i$, so $e = 2\sigma$ for an odd alternating support. Conversely, given an odd alternating support, put $s = S\sigma$. If its initial sign is positive, then $\sum_j \sigma_j = 1$ and

$$s_i = 2\sum_{j \leq i} \sigma_j - \sum_j \sigma_j \in \{-1, 1\};$$

the negative initial sign gives the negative full sign vector. Since $(1-X)S = 2$, this construction satisfies $(1-X)s = 2\sigma$, proving that every such support arises. The local field of BX is B at non-walls and $-B$ at walls. Since the perturbation field has absolute value at most $R_H < B$, the minimum is always on a wall. Therefore

$$\Psi(K) = -B + 2\Theta(G). \tag{2}$$

The identities $(1-X)S = 2$ and $XS = S - 2$ show

$$SR_n = 2R_n + \mathbb{Z}S = \{y : \text{all coefficients have the same parity}\}.$$

Its infinity-norm-one vectors are exactly all full sign vectors, and any other nonzero vector has infinity norm at least two. Let

$$R_K = \|K\|_1, \quad M = 3R_K + 1, \quad f = (M-K)S, \quad r = M + B - 2\tau.$$

Multiplication by $M - K$ satisfies

$$\|(M - K)y\|_\infty \geq (M - R_K)\|y\|_\infty.$$

On a full sign vector, its norm is $M - \min_i s_i(Ks)_i \leq M + R_K$. All other nonzero $y \in SR_n$ give norm at least $2(M - R_K) > M + R_K$. Thus

$$\lambda_{1,\infty}(fR_n) = M - \Psi(K) = M + B - 2\Theta(G). \tag{3}$$

Equations (1)–(3) give precisely the required SVP threshold equivalence. In particular $f \neq 0$. The identities themselves hold without a promise on the intermediate local-field threshold. The chosen clock amplitude makes the output radius positive, since $r = M + R_H + 1$. In a NO instance, the integer $\Theta(G)$ is at most $\tau - 1$, so equation (3) gives $\lambda_{1,\infty}(fR_n) \geq r + 2$. This is an additive gap only and does not imply a constant-factor hardness result.

All integers have polynomial encoding length. The labels are polynomially bounded, n is polynomial in the source size, and the largest guard coefficient 5^{s-1} has $O(n)$ bits. All subsequent sums, products, and norm bounds retain polynomial bit length. Ring generators and their negacyclic multiplication bases are therefore constructible in polynomial time. Finally, membership in NP follows by providing a nonzero integral coefficient vector v with $|v_i| \leq r$, whose encoding has $O(n \log(r + 1))$ bits. The multiplication matrix m_f is nonsingular; rational linear algebra computes $m_f^{-1}v$ and checks that its entries are integral. This verifies ideal membership in polynomial time and completes the proof of the theorem.

1.6 A concrete obstruction to reusing the layer at finite exponents

The failure to infer finite-p hardness has a direct short-vector explanation. Keep $f = (M-K)S$, $R_K = \|K\|_1$, and $M = 3R_K + 1$. For every fixed real $1 \leq p < \infty$ and $n \geq 4^p$, every full sign vector satisfies

$$\|(M - K)s\|_p \geq (M - R_K)n^{1/p} \geq 4(M - R_K) > 2(M + R_K).$$

Here convolution has operator norm at most R_K on coefficient ℓ_p. In contrast, $2 \in SR_n$ gives the nonzero ideal vector

$$2(M - K) \in fR_n, \qquad \|2(M - K)\|_p \leq 2(M + R_K).$$

Thus no image of the full sign layer can be a finite-p shortest vector in these dimensions. In particular, for $p = 2$ the obstruction already applies when $n \geq 16$. This rules out the direct reuse of this specific minimum-layer argument; it does not give an algorithm for arbitrary ideals or rule out a different finite-p reduction.